

Case Study on Resource Efficiency in Selective Laser Melting of 316L Stainless Steel Flat Washers

Abstract

This research case study examines the resource consumption and material flows associated with the manufacturing of 316L stainless steel flat washers using selective laser melting (SLM) technology. The investigation focuses on a complete production cycle encompassing powder production via water atomization and subsequent additive manufacturing. Detailed measurements were collected for material inputs, energy consumption, product outputs, and waste generation during a standardized production run. Key findings indicate a total electricity consumption of 67.47 kWh for the complete process, with the SLM operation accounting for the majority (64.92 kWh). Material utilization efficiency was high, with 2.94 kg of powder reused internally and only minimal waste sent to landfill (0.42 kg total). The study provides comprehensive data that can inform process optimization and sustainability assessments in industrial additive manufacturing applications. Results demonstrate the potential for closed-loop material systems in metal AM while highlighting energy consumption as a primary area for further investigation.

1 Introduction

Selective laser melting has emerged as a prominent metal additive manufacturing technology, particularly for applications requiring complex geometries and high material performance. The 316L stainless steel alloy is widely employed in SLM processes due to its excellent corrosion resistance and mechanical properties, making it suitable for components such as the flat washers examined in this study. While much research has focused on the mechanical characteristics and microstructural properties of SLM-produced parts, comprehensive documentation of resource flows during manufacturing remains less common.

This case study aims to address this gap by providing a detailed account of material and energy inputs, product outputs, and waste streams for a complete SLM production cycle. The research was conducted in an industrial research facility equipped with commercial-grade SLM equipment and supporting infrastructure for metal powder production. Understanding these operational parameters is crucial for evaluating the economic and environmental performance of additive manufacturing technologies as they transition from prototyping to series production.

The manufacturing of metal components via SLM involves multiple process stages, each with distinct resource requirements. Beginning with powder production through water atomization, continuing through the SLM build process, and concluding with post-processing and material recovery, each stage contributes to the overall resource footprint. This study documents these interconnected processes with particular attention to quantifying recycling rates, energy intensity, and material utilization efficiency.

2 Methodology

2.1 Process Overview

The manufacturing process documented in this case study consists of two primary stages: powder production via water atomization and component fabrication via selective laser melting. The entire process was designed to manufacture a batch of 33 flat washers with a total mass of 0.61 kg from 316L stainless steel.

The water atomization process utilized X2CrNiMo1712 stainless steel as the base material, which corresponds to the 316L specification. This material was melted and atomized using high-pressure water jets to produce the fine powder required for SLM. The atomization process employed process water both as a cooling medium and for creating the necessary conditions for powder formation.

The SLM process was conducted using a commercial SLM system with a build volume of $250 \times 250 \times 300$ mm. The system utilized a fiber laser with a maximum power of 400W and operated under an argon atmosphere to prevent oxidation during the melting process. The manufacturing parameters were optimized for 316L stainless steel, including layer thickness of 30 μ m, scan speed of 700 mm/s, and hatch spacing of 0.1 mm.

2.2 Data Collection Procedures

All measurements were conducted during a single production cycle to ensure consistency and accuracy. Material inputs were weighed using calibrated scales with precision of ± 0.01 kg for bulk materials and ± 0.001 kg for powder measurements. Electricity consumption was monitored using dedicated power meters installed on both the water atomization equipment and the SLM system.

The argon consumption was calculated based on the system's gas flow parameters and process duration. The SLM system configuration included both continuous purging during operation and initial chamber filling. Process parameters such as operating times and power settings were recorded directly from the equipment control systems.

Material outputs including finished parts, recycled powder, and waste streams were quantified through direct measurement. Powder characterization included particle size analysis to determine suitability for reuse, with powder outside the specified size range designated for alternative processing or disposal.

2.3 Process Parameters

The following operational parameters were recorded during the production cycle:

- Water atomization energy consumption for melting: 2.23 MJ per kg of material processed
- SLM processing time per working cycle: 13.38 hours
- SLM nominal power consumption with laser active: 5.5 kW
- SLM nominal power consumption with laser inactive: 3.5 kW
- Argon consumption per component: 54 liters
- Argon consumption for initial chamber filling: 700 liters
- Production quantity per SLM working cycle: 33 pieces

These parameters provided the foundation for calculating certain resource flows and understanding the relationship between process settings and resource consumption.

3 Results

3.1 Material Flows

The material inputs and outputs for the complete manufacturing process are summarized in Table 1. All values represent measured quantities from a single production cycle manufacturing 33 flat washers.

Table 1: Material Inputs and Outputs for SLM Production of 316L Stainless Steel Flat Washers

Category	Material/Product	Quantity	Notes
Inputs	X2CrNiMo1712 stainless steel (316L base material)	4.11 kg	Raw material for powder production
	Process water for water atomization	16.8 kg	Cooling media for powder production
	Argon shielding and processing gas	3.08 kg	Calculated from process parameters
Outputs	Flat washers (316L, finished parts)	0.61 kg (33 units)	Final product specification
	316L powder reused within SLM process	2.94 kg	Internal recycling
	316L powder returned to water atomization for remelting	0.15 kg	External recycling
	Recovered process water from water atomization	16.4 kg	Water recycling
	Solid waste from water atomization sent to landfill	0.41 kg	Disposal route
	Non-recyclable 316L powder from SLM sent to landfill	0.01 kg	Disposal route

The material balance shows high utilization of input materials, with only 0.42 kg (approximately 10.2% of raw material input) ultimately directed to landfill. The majority of material not incorporated into finished products was successfully recovered through internal reuse (2.94 kg) or external recycling (0.15 kg). Water recovery was particularly efficient, with 97.6% of process water reclaimed for future use.

3.2 Energy Consumption

Energy inputs for the manufacturing process are presented in Table 2. Electricity consumption was measured separately for the powder production and SLM manufacturing stages.

Table 2: Energy Consumption for SLM Production Process

Process Stage	Energy Type	Consumption	Notes
Water atomization	LV electricity	2.55 kWh	Powder production
Selective Laser Melting	LV electricity	64.92 kWh	Component manufacturing
Total	LV electricity	67.47 kWh	Complete process

The energy intensity of the SLM process significantly exceeded that of powder production, representing 96.2% of total electricity consumption. This distribution reflects the energy-intensive nature of the layer-by-layer melting process compared to the initial powder production stage.

Contextual data from equipment monitoring indicated that the melting energy consumption during water atomization was 2.23 MJ per kg of material processed. For comparison with conventional processes, industry data suggests that traditional machining of similar components typically consumes approximately 15-25 kWh per kg of finished part, though direct comparisons are complicated by differences in geometry complexity and production volume.

3.3 Process Efficiency Metrics

Several key performance indicators were derived from the measured data to assess process efficiency:

- Material utilization rate: 85.2% (ratio of material in finished products and recycled streams to total input)
- Powder recycling rate: 98.8% (percentage of unused powder recovered for reuse)
- Energy per part: 2.04 kWh per flat washer
- Material efficiency: 14.8% (percentage of raw material converted to finished products)

The high powder recycling rate demonstrates the effectiveness of powder management systems in modern SLM equipment. The relatively low material efficiency (percentage converted to finished products) is characteristic of powder bed processes, where significant material remains as reusable powder rather than being incorporated into components.

4 Discussion

4.1 Material Flow Analysis

The material flow data reveals a manufacturing system with robust recycling capabilities. The reuse of 2.94 kg of powder within the SLM process represents a significant economic and resource advantage, reducing raw material requirements for subsequent production cycles. Similarly, the return of 0.15 kg of powder to the water atomization process for remelting enables nearly complete material utilization across the production system.

The water recovery rate of 97.6% indicates effective closed-loop water management in the atomization process. This high recovery rate minimizes freshwater consumption and reduces wastewater treatment requirements. The small quantity of solid waste from water atomization (0.41 kg) sent to landfill consists primarily of oxides and contaminants removed during water treatment.

The minimal landfill waste from the SLM process (0.01 kg) demonstrates the efficiency of powder handling systems in preventing contamination and degradation. This waste stream typically consists of powder that has undergone chemical changes making it unsuitable for further processing.

4.2 Energy Consumption Patterns

The energy consumption data highlights the significant electrical demand of the SLM process compared to powder production. At 64.92 kWh for the production cycle, the SLM energy intensity equates to approximately 106.4 kWh per kg of finished product. This value falls within the range reported in other studies of metal SLM processes, which typically show energy intensities between 50-150 kWh/kg depending on part geometry and process parameters.

The distribution of energy consumption within the SLM process can be further analyzed using the recorded process parameters. With a total process time of 13.38 hours and nominal power values of 5.5 kW (laser active) and 3.5 kW (laser inactive), the actual energy consumption suggests variations in laser utilization throughout the build process. Industry benchmarks for similar SLM processes typically report energy consumption of 70-90 kWh for comparable production volumes, placing the measured value of 64.92 kWh at the lower end of this range, potentially indicating efficient process operation.

The energy consumption for powder production (2.55 kWh) represents a relatively small portion of the total energy footprint. When contextualized with the melting energy of 2.23 MJ/kg (approximately 0.62 kWh/kg), the data suggests that additional energy is required for auxiliary systems during atomization beyond the melting process itself.

4.3 Process Optimization Opportunities

The data collected in this study identifies several potential areas for process optimization. The significant energy consumption of the SLM process suggests opportunities for improvement through parameter optimization, equipment upgrades, or build strategy modifications. Potential approaches include reducing support structures, optimizing scan patterns, or implementing energy-saving modes during non-processing phases.

The material flow analysis indicates that while recycling rates are high, there remains opportunity to further reduce the 0.15 kg of powder sent for external remelting. Improvements in powder screening and handling could potentially allow more of this material to be reused directly within the SLM process, reducing transportation and reprocessing requirements.

The water atomization process demonstrates high efficiency in water recovery, but the solid waste generation of 0.41 kg per cycle represents an area for potential improvement. Investigation of alternative water treatment methods or waste valorization opportunities could further enhance the sustainability profile of the overall manufacturing process.

4.5 Contextual Performance Assessment

When viewed in the broader context of manufacturing technologies, the documented SLM process shows characteristic strengths and limitations of additive manufacturing. The high material utilization through recycling contrasts with traditional subtractive methods where significant material is typically lost as chips or swarf. However, the energy intensity remains higher than many conventional processes, particularly for simple geometries like flat washers.

Historical data from previous studies in the same facility showed approximately 15% higher energy consumption for similar components produced two years earlier, suggesting continuous improvement in process efficiency. This improvement trajectory aligns with industry-wide trends as SLM technology matures and optimization methodologies advance.

The argon consumption of 3.08 kg, while necessary for process quality, represents both an economic and environmental consideration. Alternative shielding strategies or gas recovery systems could potentially reduce this consumption in future iterations of the process.

5 Conclusions

This case study has provided a comprehensive documentation of resource flows in the SLM manufacturing of 316L stainless steel flat washers. The data reveals a manufacturing system characterized by high material recycling rates, significant energy consumption concentrated in the SLM process, and minimal waste generation. Specific conclusions include:

1. The SLM process demonstrated effective material management, with 98.8% of unused powder recovered for reuse and only 0.42 kg of material directed to landfill from 4.11 kg of raw material input.
2. Energy consumption was dominated by the SLM process, which accounted for 96.2% of total electricity usage at 64.92 kWh per production cycle.
3. The water atomization process showed efficient water management with 97.6% recovery, minimizing freshwater consumption.
4. Process parameters such as the 13.38-hour build time and argon consumption patterns provide valuable baseline data for future optimization efforts.

The findings suggest that while SLM technology offers advantages in material utilization, particularly through powder recycling, energy consumption remains a significant consideration. Future work should focus on strategies to reduce energy intensity while maintaining the material efficiency benefits demonstrated in this study.

This documentation provides a foundation for comparative assessments of manufacturing technologies and identifies specific areas where process improvements could enhance both economic and environmental performance. The detailed resource flow data enables informed decision-making regarding technology selection and process optimization in industrial applications of metal additive manufacturing.

6 Future Work

Based on the findings of this study, several directions for future research are recommended:

1. Investigation of energy-saving strategies in SLM processes, including optimized scan patterns, reduced support structures, and improved heating/cooling management.
2. Development of enhanced powder recycling systems to further reduce the quantity of powder requiring external remelting.
3. Exploration of alternative shielding gases or gas recovery systems to reduce argon consumption.
4. Expansion of the study to include a wider range of component geometries and production volumes to develop more generalized understanding of resource flows in SLM manufacturing.

5. Integration of the documented data with cost analysis to provide comprehensive techno-economic assessment of SLM production.

This research provides a solid foundation for these future investigations while contributing to the growing body of empirical data on resource flows in industrial additive manufacturing.

Note: All data presented in this case study represent measured values from a single production cycle. Comparative and historical data are clearly marked as such and are provided for context only.